

CHAPTER FIVE

HYPOTHESIS TESTING

INTRODUCTION

In statistics, as in life, nothing is as certain as the presence of uncertainty. However, just because we are not 100% sure of something, that is no reason why we cannot reach same conclusions that are highly likely to be true. Therefore, this chapter is aimed on the examination of the very important process of reaching conclusions based on sample information.

Objectives of the chapter

Up on completing reading this chapter, you should be able to:

- Describe the meaning of a null and an alternative hypothesis.
- Transform a verbal statement in to appropriate null and alternative hypotheses, including the determination of whether a two-tail test or a one-tail test is appropriate.
- Describe what is meant by Type I and Type II errors, and explain how these can be reduced in hypothesis testing.
- Carryout a hypothesis test for a population mean or population proportions, interpret the results of the test, and determine the appropriate business decision that should be made.

5.1 THE BASIC CONCEPTS OF HYPOTHESIS TESTING

OVERVIEW

In this section, the major concepts of hypothesis testing will be discussed and mainly focused on the concepts which include; null and alternative hypotheses, directional and non directional testing, errors in hypothesis testing and the basic procedures in hypothesis testing.

Objectives

After completing this section, you should be able to:

- Distinguish between null and alternative hypotheses.
- Explain the errors in hypothesis testing
- Describe the basic steps/procedures involved in hypothesis testing.

5.1.1 Null and Alternative Hypothesis

The first step in examining claims is to form a null hypothesis, expressed as H_0 (“H sub naught”). The null hypothesis is a statement about the value of a population parameter and is put for testing in the face of numerical evidence. The null hypothesis is either rejected or fails to be rejected. In the philosophy of hypothesis testing, the null hypothesis is assumed to be true unless we have statistically overwhelming evidence to the contrary.

Alternative hypothesis, H_1 (“H sub one”), is an assertion that holds if the null hypothesis is false. For a given test, the null and alternative hypotheses include all possible values of the Population parameter, so either one or the other must be false. Described in terms of an (unknown) population mean (μ) they might be listed as follows:

Null Hypothesis	Alternative Hypothesis
$H_0: \mu = \$10$	$H_1: \mu \neq \$10$ (μ is \$ 10, or it is not)
$H_0: \mu \geq \$10$	$H_1: \mu < \$10$ (μ is at least \$ 10, or it is less)
$H_0: \mu \leq \$10$	$H_1: \mu > \$10$ (μ is not less than \$ 10, or it is more)

Notice that each null hypothesis has an equality term in its statement (i.e., “=,” “ $\geq$,” “ $\leq$ ”). Thus, an actual population mean of \$ 10 would cause all three of them to be true. However, this does not make the three sets interchangeable. The null and alternative hypotheses are the foundation for a hypothesis test, and the selection of one of these three sets will depend on (1) the directionality or non directionality of the original claim or assertion that led to the test and (2) the purpose for which the test is being conducted.

5.1.2 Directional and Non Directional Testing

A directional claim or assertion holds that a population parameter is greater than ($>$), at least ($\geq$), no more than ($\leq$), or less than ($<$) some quantity.

A non directional claim or assertion states that a parameter is equal to some quantity. For example Ato Kebede claims that 35% of his transit riders are senior citizens.

Directional assertions lead to what are called one-tail tests, where a null hypothesis can be rejected by an extreme result in one direction only. A non-directional assertion involves a two – tail test, in which a null hypothesis can be rejected by an extreme result occurring in either direction.

5.1.3 Errors in Hypothesis Testing

Whenever we reject a null hypothesis, there is a chance that we have made a mistake i.e., that we have rejected a true statement. Rejecting a true null hypothesis is referred to as a Type I error, and our probability of making such an error is represented by the Greek letter alpha (α). This probability, which is referred to as the significance level of the test, is of primary concern in hypothesis testing.

On the other hand, we can also make the mistake of failing to reject a false null hypothesis-this is a Type II error. Our probability of making it is represented by the Greek letter beta (β).

Table 6.1 below gives the summary of the possibilities for mistakes and correct decisions in hypothesis testing. The probability of incorrectly rejecting a true null hypothesis is α , the significance level. The probability that the test will correctly reject a false null hypothesis is $(1 - \beta)$, the power of the test.

Table 6.1

The null Hypothesis (H_0) is really	
	True False
“Do not Reject H_0 ”	Correct decision Incorrect decision (Type II Error). Probability of making this error is β .
“Hypothesis test stays”	
“Reject H_0 ”	Incorrect decision (Type I error). Probability of making this error Correct decision. Probability ($1 - \beta$) is power of the test.

5.1.4 A General Procedure for Hypothesis Testing

As mentioned before, to test the validity of the claim or assumption about the population parameter, a sample is drawn from the population and analyzed. The results of the analysis are used to decide whether the claim is true or not. The steps of general procedures for any hypothesis testing are summarized below:

Step₁: State the null hypothesis (H_0) and alternative hypothesis (H_1).

The null hypothesis H_0 refers to a hypothesized numerical value or range of values of the population parameter. Theoretically hypothesis testing requires that the null hypothesis be considered true (Status quo or no difference) until it is proved false on the basis of results observed from the sample data. The null hypothesis is always expressed in the form of an equation making a claim regarding the specific value of the population parameter. That is:

$H_0: \mu = \mu_0$, where μ is population mean and μ_0 represents hypothesized parameter value.

An alternative hypothesis, H_1 , is the logical opposite of the null hypothesis, that is, an alternative hypothesis must be true when the null hypothesis is found to be false. In other words, the alternative hypothesis states that specific population parameter value is not equal to the value stated in the null hypothesis and is written as

$H_1: \mu \neq \mu_0$

Consequently $H_0: \mu < \mu_0$ or $H_1: \mu > \mu_0$

Step₂: State the level of significance, α (alpha) for the test.

The level of significance, usually denoted by α (alpha), is specified before the samples are drawn, so that the results obtained should not influence the choice of the decision maker. It is specified in terms of the level of probability of null hypothesis H_0 being wrong. In other words, a probability which has a null hypothesis may be rejected when it is true.

Step₃: Establish critical or rejection region

The sample space of the experiment which corresponds to the area under the sampling distribution curve of the test statistic is divided into the mutually exclusive regions which are called the acceptance region and the rejection or critical region.

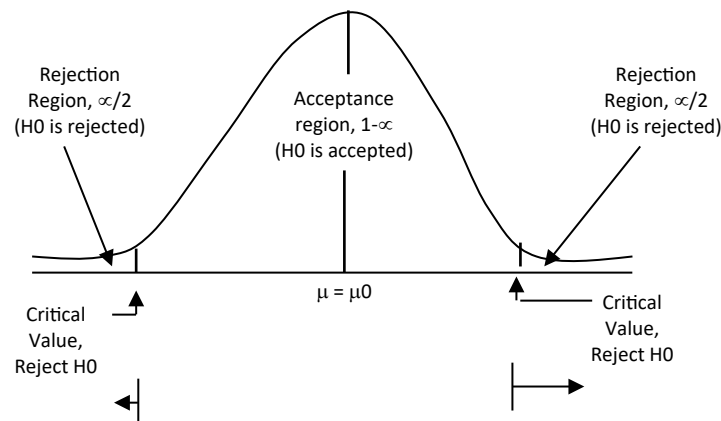


Figure 5.1 Areas of Acceptance and Rejection of H_0 (two – Tailed Test)

If the value of the test statistic falls into the acceptance region, the null hypothesis is accepted, otherwise it is rejected. The rejection region consists of all values of the test statistic that are likely to occur if null hypothesis is true.

Step 4: Calculate the suitable test statistic

The value of test statistic is calculated from the distribution of sample statistic by using the following formula

$$\text{Test statistic} = \frac{\text{Value of sample Statistic} - \text{Value of hypothesized population}}{\text{Standard error of the sample statistic}}$$

For Example, $Z = \frac{\bar{x} - \mu_0}{\frac{\delta}{\sqrt{n}}} = \frac{\bar{x} - \mu_0}{\delta / \sqrt{n}}; \delta \text{ is known } n > 30$

The choice of a probability distribution of a sample statistic is guided by the sample size n and the value of population standard deviation δ

Table 5.2: Choice of Probability distribution

Sample size n	Population standard deviation δ	
	Known	Unknown
• $n > 30$	Normal distribution	Normal distribution

• $n < 30$, population being assumed normal	Normal distribution	t-distribution
---	---------------------	----------------

Step 5: Reach a Conclusion

Compare the calculated value of the test statistic with the critical value (also called standard table value of test statistic). The decision rules for null hypothesis are as follows:

- $|z|_{\text{calculated}} \geq |z|_{\text{Table}}$; reject H_0
- $|z|_{\text{calculated}} < |z|_{\text{Table}}$; accept H_0

5.1.5 One Tailed and Two Tailed Tests

There are two types of tests of hypothesis, referred to as the one-tailed and two-tailed tests. They differ depending on the way the hypotheses are formulated.

i) Null hypothesis and alternative hypothesis stated as

$$H_0: \mu = \mu_0 \text{ and } H_1: \mu \neq \mu_0$$

Imply that any deviation (either on the lower side or higher side) of the calculated value of test statistic from a hypothesized value μ_0 leads to rejection of the null hypothesis. Hence, it is necessary to keep the rejection region on 'both tails' of the sampling distribution of the test statistic. This type of test is called two-tailed test.

A summary of certain critical values at various significance levels for test statistic Z is given in table 5.3.

Table 5.3: Summary of certain critical values for sample statistic z

Rejection Region	Level of significance, per cent				
	10%	5%	1%	0.5%	0.02%
One –tailed region	1.28	1.64	2.33	2.58	2.88
Two-tailed region	1.645	1.96	2.58	2.81	3.08

5.2 HYPOTHESIS TESTING FOR POPULATION PARAMETERS WITH LARGE SAMPLES

Dear learner, notice that, hypothesis testing involving large samples (sample size, $n > 30$) is based on the assumption that the population from which the sample is drawn has a normal distribution. Consequently the sample distribution of mean $\bar{x}$ is also normal. Even if the population does not have a normal distribution, the sampling distribution of mean $\bar{x}$ is assumed to be normal due to the central limit theorem because the sample size is large.

Objectives of the section

After completing this section, the learner will be able to:

- Distinguish between hypothesis testing under single population and two population
- Differentiate between one-tailed and two tailed hypothesis testing based on examples
- Appreciate the application of hypothesis testing in evaluating the difference of two population means

5.2.1 Hypothesis Testing for Single Population Mean

a) Let μ_0 be hypothesized value of the population mean to be tested. For this, the null and alternative hypotheses for two tailed test are defined as:

$$H_0: \mu = \mu_0 \text{ or } \mu - \mu_0 = 0 \text{ (Two- tailed test)}$$

$$H_1: \mu \neq \mu_0$$

If standard deviation δ of the population is known, then based on the central limit theorem, the sampling distribution of mean $\bar{x}$ would follow the standard normal distribution for a large sample size. The Z test statistic is given by:

$$Z = \frac{\bar{x} - \mu}{\frac{\delta}{\sqrt{n}}} = \frac{\bar{x} - \mu}{\delta / \sqrt{n}}$$

In this formula, the numerator $\bar{x} - \mu$, measures how far (in an absolute sense) the observed sample mean $\bar{x}$ is from the hypothesized mean (μ). The denominator $\frac{\sigma}{\sqrt{n}}$ is the standard error of the mean, so the Z-test statistic represents how many standard errors $\bar{x}$ is from (μ).

If the population standard deviation σ is not known, then a sample standard deviation (s) is used to estimate (σ). The value of the test statistic is

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Since the rejection region is divided in to two equal parts of $Z_{\alpha/2}$ each at the two tails of a normal curve, therefore the decision rule for the two – tailed test would be

- Reject H_0 if $Z_{\text{cal}} < -Z_{\alpha/2}$ or $Z_{\text{cal}} > Z_{\alpha/2}$
- Otherwise accept H_0 ,

Where $Z_{\alpha/2}$ is the table value (also called critical value) of Z at a chosen level of significance α .

b) Large sample ($n > 30$) hypothesis testing about a population mean for a left-tailed test is of the form

$H_0: \mu \geq \mu_0$ and $H_1: \mu < \mu_0$ (left- tailed test)

$$\text{Test statistic: } Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Decision rule: Reject H_0 if $Z_{\text{cal}} < -Z_{\alpha}$ (Table value of Z at α)

Otherwise accept H_0

C) Large sample ($n > 30$) hypothesis testing about a population mean for a right – tailed test is of the form:

$H_0: \mu \leq \mu_0$ and $H_1: \mu > \mu_0$ (Right–tailed test)

$$\text{Test statistic: } Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Decision rule: Reject H_0 if $Z_{\text{cal}} > Z_{\alpha}$ (Table value of Z at α)

Otherwise accept H_0

Example 1: Individual filing of income tax returns prior to 30 June had an average refund of \$ 1200. Consider the population of Last “minute” filers

who file their returns during the last week of June. For a sample of 400 individuals who filed a return between 25 and 30 June, the sample mean refund was \$1054 and the sample standard deviation was \$1600. Using 5 per cent level of significance, test the belief that the individuals wait until the last week of June to file their returns to get a higher refund than early filers.

Solution: Let us take the null hypothesis H_0 that the individuals wait until the last week of June to file their returns get a higher refund than early filers, that is

$H_0: \mu \geq 1200$ and $H_1: \mu < 1200$ (left – tailed test)

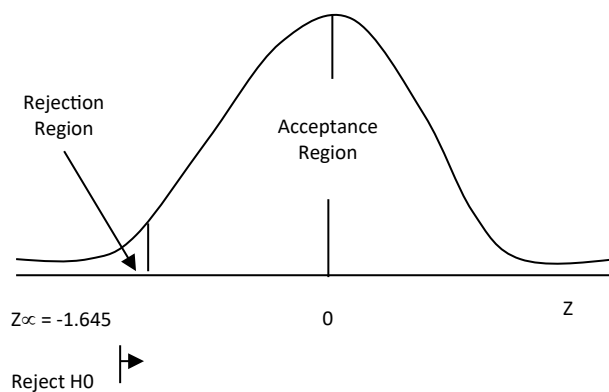


Figure 5.2

Given $n = 400$, $\delta = 1600$, $\bar{x} = 1054$, $\alpha = 5\%$. Thus using the test statistic,

$$Z = \frac{\bar{x} - \mu}{\delta / \sqrt{n}} = \frac{1054 - 1200}{1600 / \sqrt{400}} = -\frac{146}{80} = -1.825$$

Since the calculated value $Z_{\text{cal}} = -1.825$ is less than its critical value $Z_{\alpha} = -1.645$ at $\alpha = 5\%$ level of significance, the null hypothesis is rejected. Hence we conclude that individuals who wait until the last week of June are likely to receive a refund of less than \$ 1200.

Example 2: A packaging device is set to fill detergent powder packets with a mean weight of 5 kg. The standard deviation is known to be 0.01 kg. These are known to drift upwards over a period of time due to machine fault, which is not tolerable, A random sample of 100 packets is taken and weighed.

This sample has a mean weight of 5.03 kg and a standard deviation of 0.21kg. Can we conclude that the mean weight produced by the machine has increased? Use a 5 percent level of significance.

Solution: Let us take the null hypothesis H_0 that mean weight has increased, that is

$$H_0: \mu=5 \text{ and } H_1: \mu > 5 \quad (\text{right - tailed test})$$

Given $n = 100$, $\bar{x} = 5.03$ kg, $S = 0.21$ kg and $\alpha = 5\%$. Thus using the test statistic,

$$Z = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{5.03 - 5}{0.21 / \sqrt{100}} = \frac{0.03}{0.021} = 1.428$$

Since calculated value $Z_{\text{cal}} = 1.428$ is less than its critical value $Z_{\alpha} = 1.645$, the H_0 is accepted as shown by fig. 5.3. Hence we conclude that mean weight is likely to be more than 5 kg.

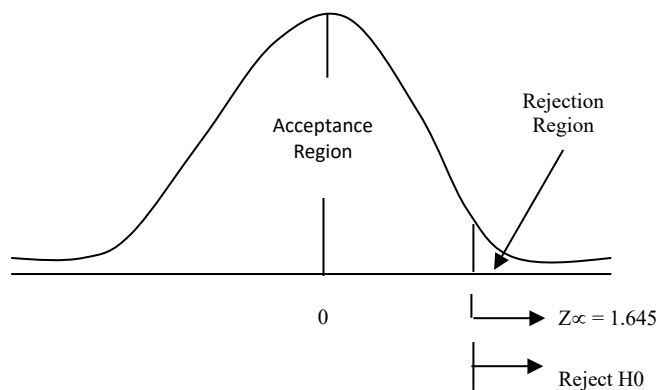


Figure 5.3

5.2.1.1 One –Tail Testing of Mean, δ known

Example: The light bulbs in an industrial ware house have been found to have mean lifetime of 1030 hours, with a standard deviation of 90 hours. The warehouse manager has been approached by a representative of Extendabulb, a company that makes a device intended to increase bulb life. The manager is concerned that the average life time of Extendbulb-equipped bulbs might not be any greater than the 1030 hours historically experienced. In a subsequent test, the manager tests 40 bulbs equipped

with the device e and find their mean life to be 1061.6 hours. Does extend bulb relay work?

i) Formulate the null and Alternative hypotheses

The warehouse manager's concern that Extendabulb equipped bulbs might not be any better than those used in the past leads to a directional test. Accordingly, the null and alternative hypotheses are:

$H_0: \mu \leq 1030$ hours Extendabulb is no better than the present system.

$H_1: \mu > 1030$ hours Extendabulb really does increase bulb life.

At the center of hypothesized distribution will be the highest possible value for which H_0 could be true, $\mu = 1030$ hours.

ii) Select the Significance Level (α)

The level chosen for the test will be $\alpha = 0.05$. If Extendabulb really has no favorable effect, the maximum probability of our mistakenly concluding that it does will be 0.05.

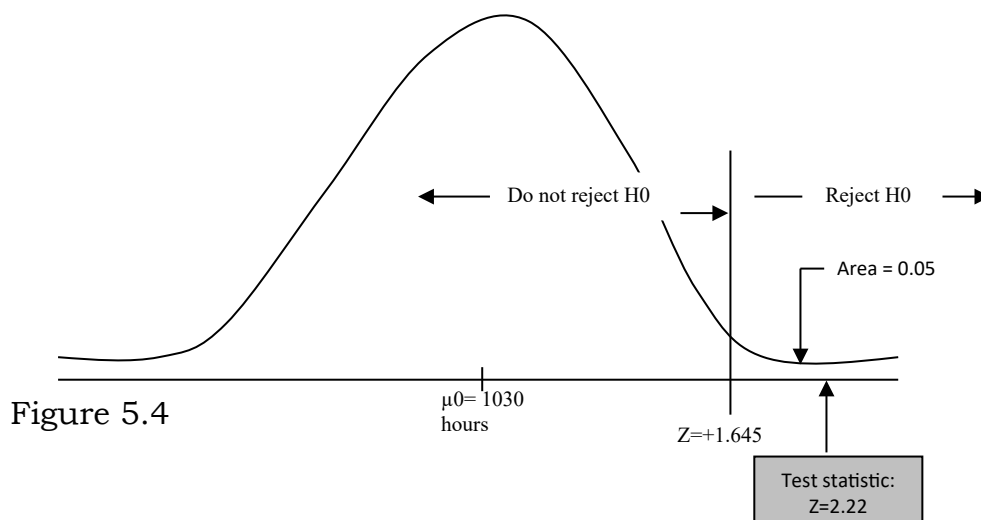
iii) Select the Test Statistic and Calculate Its Value

As in the previous test, the population standard deviation (σ) is known and the sample size is large, so the normal distribution is appropriate and the test statistic will be Z. It is calculated as

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{1061.6 - 1030.0}{90 / \sqrt{40}} = 2.22$$

iv) Select the Critical Value for the Test Statistic and State the Decision Rule

For a right-tail Z-test in which $\alpha = 0.05$, $Z = 1.645$ will be the boundary separating the non rejection and rejection regions. This critical value for the test is included in Figure 5.4. The decision rule can be stated as "Reject H_0 if calculated $Z > 1.645$, otherwise do not reject (accept)."



v) Compare calculated and Critical values and reach a conclusion for the Null hypothesis

The calculated value, $Z = 2.22$, falls within the rejection region of the diagram in figure 5.4. At the 0.05 level of significance, the null hypothesis is rejected.

vi) Make the Related Business Decision

The results suggest that Extendabulb does increase the mean lifetime of the bulb. The difference between the mean of the hypothesized distribution, $\mu_0 = 1030$ hours and the observed sample mean, $\bar{x} = 1061.6$, is judged too great to have occurred by chance. The firm may wish to incorporate Extenda bulb into its warehouse lighting system.

5.2.3 Testing A Mean, Population Standard Deviation Unknown

The true standard deviation of a population will usually be unknown. The t-test is appropriate for hypothesis tests in which the sample standard deviation (s) is used in estimating the value of the population standard deviation, σ . The t-test is based on the t distribution (with number of degrees of freedom, $df = n-1$) and the assumption that the population is approximately normally distributed. As the sample size becomes larger, the assumption of population normality becomes less important.

When degree of freedom is small, the t distribution is flatter and more spread out than the normal distribution, but for larger degrees of freedom, successive members of the family more closely approach the normal distribution. As the number of degrees of freedom approaches infinity, the two distributions become identical.

Like the Z-test, the t-test depends on the sampling distribution for the sample mean. The appropriate test statistic is similar in appearance, but

includes S instead of σ , because S is being used to estimate the (unknown) value of σ . The test statistic can be calculated as follows:

Test statistic, t – test for a sample mean:

$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}} \text{ where } S/\sqrt{n} = \text{estimated standard error for the sample mean, } = \frac{s}{\sqrt{n}}$$

$\bar{x}$ = Sample mean

μ_0 = hypothesized population mean

n = Sample size

One – Tail Testing Of a Mean, σ Unknown

The Chekzar Rubber Company, in financial difficulties because of a poor reputation for product quality, has come out with an advertising campaign claiming that the mean lifetime for Chekzar tires is at least 60,000 miles in high way driving. Skeptical, the editors of a consumer magazine purchase 36 of the tires and test them in high way use. The mean tire life in the sample is $\bar{x} = 58,341.69$ miles, with a sample standard deviation of $S = 3,632.53$ miles.

i) Formulate the Null and Alternative Hypotheses

Because of the directional nature of the advertising claim and the editors' skepticism regarding its truthfulness, the null and alternative hypotheses are

$H_0 = \mu \geq 60,000 \text{ miles}$ *The mean tire life is 60,000 miles.*

$H_1: \mu < 60,000 \text{ miles}$ *The mean tire life is under 60,000 miles.*

ii) Identify the critical value for the Test statistic and state the decision rule.

For this test, α has been specified as 0.01. The number of degrees of freedom is $df = (n-1)$, or $(36-1) = 35$. The t distribution table is now used in finding

the value of t that corresponds to a one-tail area of 0.01 and df 35 degrees of freedom. Referring to the 0.01 column and 35th row of the t table, this critical value is found to be $t = -2.438$. (Although the value listed is positive, remember that the distribution is symmetrical, and we are looking for the left-tail boundary.) The rejection and non rejection regions are shown in Figure 5.5 and the decision rule can be stated as “Reject H_0 if the calculated t is less than -2.438 , other wise do not reject.”

iii) Compare the calculated and critical values and reach a conclusion for the Null Hypothesis

The calculated test statistic, $t = -2.739$, is less than the critical value, $t = -2.438$, and falls into the rejection region of the test. The null hypothesis, $H_0 = \mu \geq 60,000 \text{ miles}$, must be rejected.

iv) Make the related business decision

The test results support the editors’ doubts regarding Chekzar’s advertising claim. The magazine may wish to exert either readership or legal pressure on Chekzar to modify its claim.

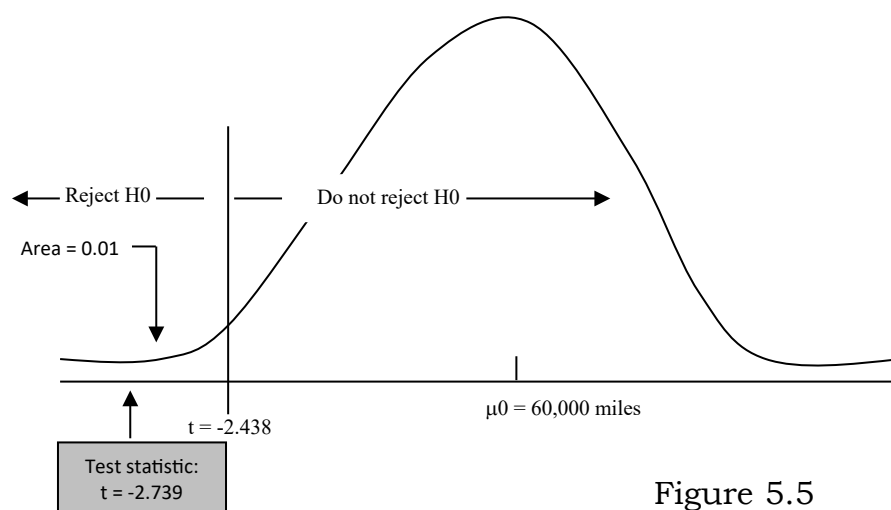


Figure 5.5

Two – Tail Testing of a Mean, Unknown

Example: The credit manager of a large department store claims that the mean balance for the store’s charge account customers is \$410. An independent auditor selects a random sample of 18 accounts and finds a mean balance of $\bar{x} = \$511.33$ and a standard deviation of $S = \$183.75$. If the

manager's claim is not supported by these data, the auditor intends to examine all charge account balances. If the population of account balance is assumed to be approximated normally distributed, what action should the auditor take?

i) Formulate the null and Alternative hypotheses.

$H_0 : \mu = \$410$ *The mean balance is actually \$410*

$H_1 : \mu \neq \$410$ *The mean balance is some other value.*

In evaluating the manager's claim, a two-tail test is appropriate since it is a non directional statement that could be rejected by an extreme result in either direction. The center of the hypothesized distribution of sample means for samples of $n = 18$ will be $\mu_0 = \$410$.

ii) Select the significance level

For this test, we will use the 0.05 level of significance. The sum of the two tail areas will be 0.05.

iii) Select the test statistic and calculate its value.

The test statistic is $t = (\bar{x} - \mu_0) / S\bar{x}$, and the t distribution will be used to describe the sampling distribution of the mean for samples of $n = 18$. The center of the distribution is $\mu = \$410$, which corresponds to $t = 0.000$. Since the population standard deviation is unknown, s is used to estimate σ . The sampling distribution has an estimated standard error of

$$S\bar{x} = \frac{S}{\sqrt{n}} = \frac{\$183.75}{\sqrt{18}} = \$43.31 \quad \text{and the calculated value of } t \text{ will be}$$

$$t = \frac{\bar{x} - \mu_0}{S\bar{x}} = \frac{\$511.33 - \$410.00}{\$43.31} = 2.340$$

iv) Identify critical values for the test statistic and state the decision rule

For this test, $\alpha = 0.05$, and the number of degrees of freedom will be $df = (n-1)$, or $(18-1) = 17$. The rejection and non rejection areas are shown in Figure 5.6, and the decision rule can be stated as "Reject H_0 if the calculated t is either < -2.110 or $> +2.110$, otherwise do not reject."

v) Compare the calculated and critical values and reach a conclusion for the null hypothesis

The calculated test statistic, $t = 2.340$, exceeds the upper boundary and falls into this rejection region. H_0 is rejected.

vi) Make the related business decision

The results suggest that the mean charge account balance is some value other than \$ 410. The auditor should proceed to examine all charge account balances.

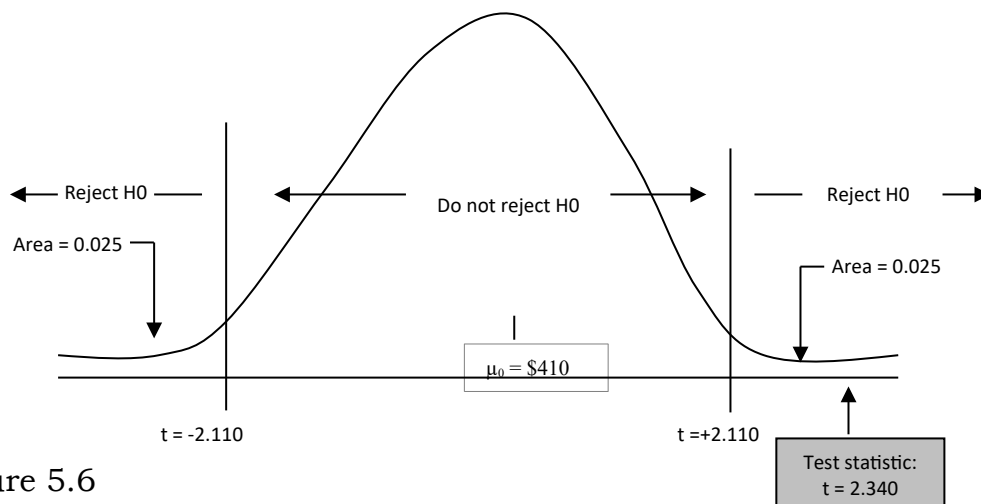


Figure 5.6

Activity 5.1

Dear learner, try to practice what we have discussed above by solving the following questions which are set based on the examples given.

1. For a simple random sample of 15 items from a population that is approximately normally distributed, $\bar{x} = 82$ and $S = 20.5$. At the 0.05 Level of significance, test $H_0: \mu \geq 90$ versus $H_1: \mu < 90$.
2. The average length of a flight by regional airlines in the United States has been reported as 231 miles. If a simple random sample of 30 flights by regional airlines were to have $\bar{x} = 246.6$ miles and $S = 42.8$ miles, would this tend to cast doubt on the reported average of 231 miles?

Use a two –tail test and the 0.05 level of significance in arriving at your answer.

5.2.4 HYPOTHESIS TESTING FOR DIFFERENCE BETWEEN MEAN VALUES OF TWO POPULATIONS

If we have two independent populations each having its mean and standard deviation as:

Population	Mean	Standard Deviation
1	μ_1	σ_1
2	μ_2	σ_2

Then we can extend the hypothesis testing concepts developed in the previous section to test whether there is any significant difference between the means of these populations.

Let two independent random samples of large size n_1 and n_2 be drawn from the first and second population, respectively. Let the sample means so calculated be $\bar{x}_1$ and $\bar{x}_2$. The Z-test statistic used to determine the difference between the population means $(\mu_1 - \mu_2)$ is based on the difference between the sample mean $(\bar{x}_1 - \bar{x}_2)$. This test statistic will follow the standard normal distribution for a large sample due to the central limit theorem. The Z-test statistic is

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\delta \bar{x}_1 - \bar{x}_2} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Where $\delta \bar{x}_1 - \bar{x}_2 =$ Standard error of the statistic $(\bar{x}_1 - \bar{x}_2)$

$\bar{x}_1 - \bar{x}_2 =$ difference between two sample means, that is, sample statistic.

$\mu_1 - \mu_2 =$ difference between population means, that is, hypothesized population parameter.

If the standard deviations σ_1 and σ_2 of each of the populations are not known, then we may estimate the standard error of sampling distribution of the sample statistic $\bar{x}_1 - \bar{x}_2$ by substituting the sample standard deviations S_1 and S_2 as estimates of the population standard deviations. Under this condition, the standard error of $\bar{x}_1 - \bar{x}_2$ is estimated as:

$$S_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

The standard error of the difference between standard deviation of sampling distribution is given by

$$\delta_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\delta_1^2}{2n_1} + \frac{\delta_2^2}{2n_2}}$$

If δ_1 and δ_2 are not known, then we use standard deviation S_1 and S_2 of the sampling distribution.

The null hypothesis that there is no difference between two population means is stated as:

$$H_0: \mu_1 = \mu_2 \text{ and } H_1: \mu_1 \neq \mu_2$$

If null hypothesis is true, then $\mu_1 - \mu_2 = 0$. The critical (or table) value of the Z-test statistic at a particular level of significance for two-tailed test is given by $\pm Z_{\alpha/2}$. For a two-tailed test, the rejection region of the null hypothesis H_0 is divided into two equal parts of $Z_{\alpha/2}$ each at the two tails of a normal curve, therefore the decision rule would be

- Reject H_0 if $Z_{\text{cal}} > Z_{\alpha/2}$ or $Z_{\text{cal}} < -Z_{\alpha/2}$
- Otherwise accept H_0

Example: A firm believes that the tires produced by process A on an average last longer than tires produced by process B. To test this belief, random samples tires produced by the two processes were tested and the results are:

Process	Sample size	Average lifetime (in km)	Standard Deviation (in km)
A	50	22,400	1000
B	50	21,800	1000

Is there evidence at a 5 per cent level of significance that the firm is correct in its belief?

Solution: Let us take the null hypothesis that there is no significant difference in the average life of tires produced by processes A and B, that is,

$$H_0: \mu_1 = \mu_2 \text{ or } \mu_1 - \mu_2 = 0 \quad \text{and} \quad H_1: \mu_1 \neq \mu_2$$

Given: $\bar{x}_1 = 22,400\text{km}$, $\bar{x}_2 = 21,800\text{km}$, $\delta_1 = \delta_2 = 1000\text{km}$ and $n_1 = n_2 = 50$.

Thus using the Z-test statistic

$$Z = \frac{(x_1 - x_2) - (\mu_1 - \mu_2)}{\delta_{\bar{x}_1 - \bar{x}_2}} = \frac{\bar{x}_1 - \bar{x}_2}{\delta_{\bar{x}_1 - \bar{x}_2}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\delta_1^2}{n_1} + \frac{\delta_2^2}{n_2}}}$$

$$Z = \frac{22,400 - 21,800}{\sqrt{\frac{(1000)^2}{50} + \frac{(1000)^2}{50}}} = \frac{600}{\sqrt{20,000 + 20,000}} = \frac{600}{200} = 3$$

Since the calculated value $Z_{\text{cal}} = 3$ is more than its critical value $Z_{\alpha/2} = \pm 1.645$ at 5 per cent level of significance, therefore H_0 is rejected. Hence we conclude that the tires produced by process A last lower than those produced by process B.

Exercise 5.1

Based on the concepts discussed so far, try to apply what you have attained on the following questions.

1. Taxco, a firm specializing in the preparation of income tax returns, claims the mean refund for customers who received refunds last year was \$150. For a random sample of 12 customers who received refund last year, the mean amount was found to be \$125, with a standard deviation of \$43. Assuming that the population is approximately normally distributed, and using the

0.10 level in a two-tail test, do these results suggest that Taxco's assertion may be accurate?

2. According to the New York Stock Exchange, the mean portfolio value for US senior citizens who are shareholders is \$ 183,000. Suppose a simple random sample of 50 senior citizen shareholders in a certain region of the United States is found to have a mean portfolio value of \$198,700 with a standard deviation of \$ 65,000. From those sample results and deviation of \$65,000 From these sample results, and using the 0.05 level of significance in a two –tail test, comment on whether the mean portfolio value for all senior citizen shareholders in this region might not be the same as the mean value reported for their counterparts a cross the nation.

5.3 HYPOTHESIS TESTING FOR POPULATION PROPORTION

Sometimes instead of testing a hypothesis pertaining to a population mean, a population proportion (P) values in a particular category is considered. For this, a random sample of size n is selected to compute the proportion of successes in the particular sample as follows:

$$P = \frac{\text{Number of successes in the Sample}}{\text{Sample size}} = \frac{x}{n}$$

Occasions may arise when we wish to compare a sample proportion, p, with a value that has been hypothesized for the population proportion π .

The theoretically correct distribution for dealing with proportions is the binomial distribution. However, the normal distribution, is good approximation when $n\pi \geq 5$ and $n(1-\pi) \geq 5$. The larger the sample size, the better this approximation becomes, and for most practical settings, this condition is satisfied. When using the normal distribution for hypothesis tests of sample proportion, the test statistics is as follows:

Test statistic, Z – test for a sample proportion:

$$Z = \frac{p - \Pi_0}{\sigma_p} \quad \text{Where } p = \text{Sample proportion}$$

Π_0 = Hypothesized population proportion

n = Sample size

σ_p = standard error of the distribution of the sample proportion

$$\sigma_p = \sqrt{\frac{\Pi_0(1 - \Pi_0)}{n}}$$

Objectives of the section

After you read this section thoroughly, you will be expected to:

- Differentiate between population proportion and sample proportion
- Appreciate the application of proportions in testing the real business hypothesis testing problems
- Distinguish the application of single population proportion and two population proportions in hypothesis testing

5.3.1 One-Tail Tasting Of a Proportion

Directional tests for a proportion are similar to the preceding example, but have only one-tail area in which the null hypothesis can be rejected.

Example: Let us assume that in an administrative decision, the ministry of health closed the cardiac surgery units of several hospitals that either performed fewer than 150 operations per year or had mortality rates higher than 5%. In one of the closed surgery units, 100 operations had been performed during the preceding year, with a mortality rate of 7%. At the 0.01 level of significance, was the mortality rate of this hospital significantly greater than the 5% cut off point? Consider the hospital's performance as representing a

sample from the population of possible operations it might have performed if the patients had been available.

i) Formulate the null and alternative hypotheses

The null hypothesis makes the assumption that the “population” mortality rate for the hospital cardiac surgery unit is really no greater than 0.05, and that the observed proportion, $P = 0.07$, was simply due the chance variation.

$H_0: \pi \leq 0.05$ the true mortality rate for the units is no more than 0.05.

$H_1: \pi > 0.05$ the true mortality rate is greater than 0.05.

The center of the hypothesized distribution, $\pi_0 = 0.05$, is the highest possible value for which the null hypothesis could be true.

ii) Select the significance level

The significance level has been specified as $\alpha = 0.01$. If the null hypothesis were really true, there would be no more than a 0.01 probability of rejecting it.

iii) Select the test statistic and calculate its value

The test statistic will be Z , calculated as $Z = (P - \pi_0) / \sigma_P$. The standard error of the sample proportion and the calculated value of the test statistic are

$$\sigma_P = \sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \sqrt{\frac{0.05(1-0.05)}{100}} = 0.02179$$

and

$$Z = \frac{P - \pi_0}{\sigma_P} = \frac{0.07 - 0.05}{0.02179} = 0.92$$

iv) Identify the critical value for the Test statistic and state the decision rule

For the 0.01 level, the critical value of Z is $Z = +2.33$. The decision rule can be stated as “Reject H_0 if calculated $Z > +2.33$, otherwise do not reject.”

v) Compare the calculated and critical values and reach a conclusion for the null hypothesis

Since the calculated value, $Z = 0.92$, is less than the critical value, it falls into the non rejection region on figure 5.7 and the null hypothesis, $H_0: \pi \leq 0.05$, cannot be rejected.

vi) Make the related business decision

The cardiac surgery mortality rate for this hospital could have been as high as 0.07 merely by chance, and closing it could not be justified strictly on the basis of a “Significantly greater than 0.05.” guideline.

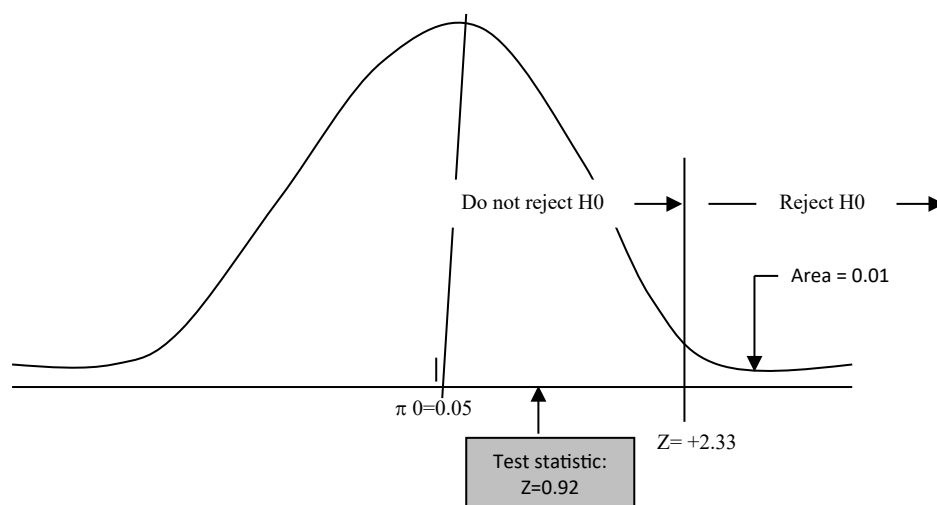


Figure 5.7

5.3.2 Two – Tail Testing Of a Proportion

Whenever the null hypothesis involves a proportion and is non directional, this technique is appropriate. To demonstrate how it works, consider the following situation.

Example: The career services director of Hobart university has said that 70% of the school's seniors enter the job market in a position directly related to their under graduate field of study. In a sample consisting of 200 of the graduates from last year's class, 66% have entered jobs related to their field of study.

i) Formulate the Null and Alternative Hypotheses

The director's statement is non directional and leads to null and alternative hypotheses of

$\mu_0 = \pi = 0.70$ the proportion of graduates entering jobs in their field is 0.70.

$H_1 = \pi \neq 0.70$ the proportion is some value other than 0.70

ii) Select the significance level

For this test, the 0.05 Level will be used. The sum of the two-tail areas will be 0.05.

iii) Select the Test statistic and calculate its value

The test statistic will be Z, the numbers of standard error units from the hypothesized population proportion, $\pi = 0.70$, to the sample proportion, $P=0.66$. The standard error of the sample proportion is

$$\sigma P = \sqrt{\frac{\pi_o(1-\pi_o)}{n}} = \sqrt{\frac{0.7(1-0.7)}{200}} = 0.0324$$

And the calculated value of Z will be

$$Z = \frac{P - \pi_o}{\sigma_p} = \frac{0.66 - 0.70}{0.0324} = -1.23$$

iv) Identify critical values for the Test statistic and state decision rule

Since the test is two-tail and the selected Level of significance is 0.05, the critical values will be $z = -1.96$ and $Z = +1.96$. The decision rule can be

stated as “Reject H_0 if calculated Z is either < -1.96 or $> + 1.96$, other wise do not reject.”

v) Compare the calculated and critical values and reach a conclusion for the Null hypothesis.

The calculated value of the test statistic, $Z = -1.23$, falls between the two critical values, placing it in the non rejection region of the distribution shown in figure 5.8. The null hypothesis is not rejected.

vi) Make the relate decision

Failure to reject the null hypothesis leads us to conclude that the proportion of graduates who enter the job market in careers related to their field of study could indeed be equal to the claimed value of 0.70. If the career services director has been making this claim to her students or their parents, this analysis would show that her assertion not be challenged.

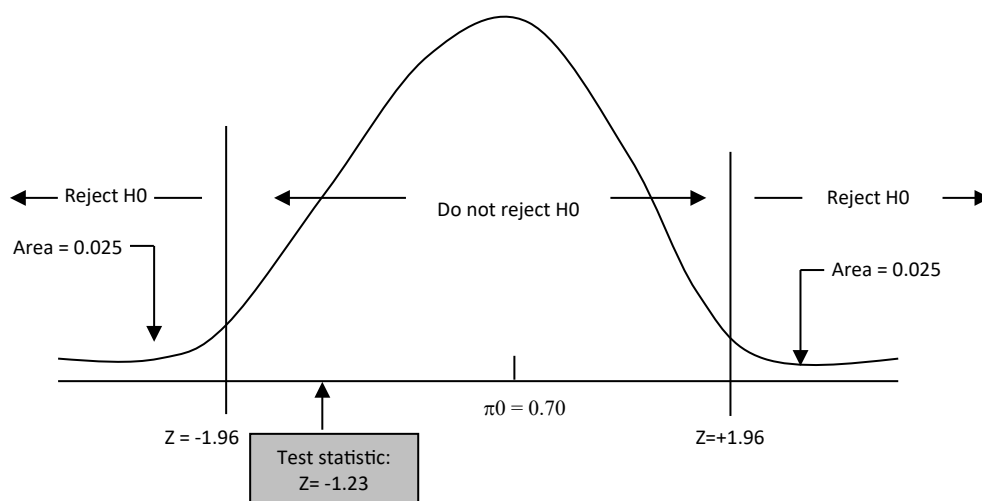


Figure 5.8

Activity 5.2

Dear student, try to practice the following questions based on what we have discussed in the above section.

1. According to the human resources director of a plant, no more than 5% of employees hired in the past year have violated their pre employment agreement of five illegal drugs. The agreement specified that random urine checks could be carried out to ascertain

compliance. In a random sample of 400 employees, screening detected at least one of these drugs in the systems of 8% of those tested. At the 0.025 level, is the human resources director's claim credible?

- 2.** In the past, 44% of those taking a public accounting qualifying exam have passed the exam on their first try. Lately, the availability of exam preparation books and tutoring sessions may have improved the likelihood of an individual's passing on his or her first try. In a sample of 250 recent applicants, 130 passed on their first attempt. At the 0.05 level of significance, can we conclude that the proportion passing on the first try has increased?

Exercise 5.2

1. For a simple random sample, $n = 1000$ and $P = 0.47$. At the 0.05 level, test $H_0: p \geq 0.50$ versus $H_1: p < 0.50$.
2. A simple random sample of 300 items is selected from a large shipment, and testing reveals that 4% of the sampled items are defective. The supplier claims that no more than 2% of the items in the shipment are defective. Carry out an appropriate hypothesis test and comment on the credibility of the supplier's claim.
3. During a year end audit, discrepancies were found in a company's invoice ledger. Consequently, the controller had all invoices for the year checked to determine if they were correctly recorded in the ledger. The proportion of incorrectly recorded invoices was found to be 0.04. The controller instituted a new procedure for processing invoices. Subsequently, a random sample of 500 invoices was checked to determine whether the proportion incorrectly recorded had changed from 0.04.
 - a) Perform a hypothesis test at the 5 per cent level if 11 of the sample invoices were recorded incorrectly.
 - b) What is implied by the test result?

5.3.3 HYPOTHESIS TESTING FOR DIFFERENCE BETWEEN PROPORTIONS OF TWO INDEPENDENT POPULATIONS

Let two independent populations each having proportion and standard deviation of an attribute be as follows:

Population	Proportion	Standard Deviation
1	P_1	σ_{P1}
2	P_2	σ_{P2}

The hypothesis testing concepts developed in the previous section can be extended to test whether there is any difference between the proportions of these populations. The null hypothesis that there is no difference between two population proportions is stated as:

$$H_0: P_1 = P_2 \text{ or } P_1 - P_2 = 0 \text{ and } H_1 = P_1 \neq P_2$$

The sampling distribution of difference in sample proportions $\bar{P}_1 - \bar{P}_2$ is based on the assumption that the difference between two population proportions, $P_1 - P_2$ is normally distributed. The standard deviation (error) of sampling distribution of $P_1 - P_2$ is given by

$$\delta_{\bar{P}_1 - \bar{P}_2} = \sqrt{\frac{P_1(1-P_1)}{n_1} + \frac{P_2(1-P_2)}{n_2}} ; \quad q_1 = 1 - p_1 \text{ and } q_2 = 1 - p_2$$

Thus the Z test statistic for the difference between two population proportions is stated as:

$$Z = \frac{(\bar{P}_1 - \bar{P}_2) - (P_1 - P_2)}{\delta_{\bar{P}_1 - \bar{P}_2}} = \frac{\bar{P}_1 - \bar{P}_2}{\delta_{\bar{P}_1 - \bar{P}_2}}$$

Invariably, the standard error δ of difference between sample proportions is not known. Thus when a null hypothesis states that there is no difference

between the population proportions, we combine two sample proportions, $\bar{p}_1$ and $\bar{p}_2$ to get one unbiased estimate of population proportion as follows:

Pooled estimate
$$\bar{p} = \frac{n_1 \bar{p}_1 + n_2 \bar{p}_2}{n_1 + n_2}$$

The z- test statistic is then restated as:

$$Z = \frac{\bar{p}_1 - \bar{p}_2}{S_{\bar{p}_1 - \bar{p}_2}}; S_{\bar{p}_1 - \bar{p}_2} = \sqrt{\bar{p}(1 - \bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

Example 1: An auditor claims that 10 per cent of customers' ledger accounts are carrying mistakes of posting and balancing a random sample of 600 was taken to test the accuracy of posting and balancing and 45 mistakes were found. Are these sample results consistent with the claim of the auditor? Use 5 per cent level of significance.

Solution: let us take the null hypothesis that the claim of the auditor is valid, that is

$H_0: P = 0.10$ and $H_1: P \neq 0.10$ (Two tailed test)

Give $\bar{p} = \frac{45}{600} = 0.075$, $n = 600$, and $\alpha 5$ percent.

Thus using the Z – test statistic

$$Z = \frac{\bar{p} - p_0}{S_{\bar{p}_1 - \bar{p}_2}} = \frac{0.075 - 0.10}{\sqrt{\frac{0.10 \times 0.90}{600}}} = \frac{-0.025}{0.0122} = -2.949$$

Since Z cal (= -2.049) is less than its table value z_c (= -1.96), null hypothesis is rejected. Hence, we conclude that the clam of the auditor is not valid.

Example 2: Random samples of 1600 workers in region 1 and 1400 workers in region 2 have been obtained to determine whether the population proportions unemployed in the two regions are different. Perform a hypothesis test at the 5 percent level if the numbers unemployed in the samples were 120 in region 1 and 84 in region 2.

Solution: The word different in the statement of the example means $P_1 \neq P_2$ or $P_1 - P_2 \neq 0$. Hence, we have the hypotheses:

$$H_0 = P_1 - P_2 = 0 \quad H_1: P_1 - P_2 \neq 0.$$

Notice: Dear learner, notice that different books may use different terms in the calculation of test statistic.

To make things clear, let us consider the terms that are used on different books and proceed with the solution:

The First Option:

1. The hypothesis

The test is two tailed, with $1/2 (0.05) = 0.025$ in each tail; because $Z_{0.025} = 1.96$, we have the decision rule.

2. The Decision Rule

Reject H_0 if / sample $Z / > 1.96$

That is, we will reject H_0 if the sample Z is less than -1.96 , or if it is greater than 1.96 .

We need $\bar{p}_c$ to use in completing $S_{\bar{p}_1 - \bar{p}_2}$. It is

$$\begin{aligned}\bar{p}_c &= \frac{\text{sum of numbers unemployed in the two samples}}{n_1 + n_2} \\ &= \frac{120 + 84}{1600 + 1400} \\ &= 0.068\end{aligned}$$

Next we compute

$$\bar{q}_c = 1 - \bar{p}_c$$

$$0.932 = 1 - 0.068$$

$$\begin{aligned}S_{\bar{p}_1 - \bar{p}_2} &= \sqrt{\bar{p}_c \bar{q}_c \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \\ &= \sqrt{(0.068)(0.932) \left(\frac{1}{1600} + \frac{1}{1400} \right)} \\ &= 0.00921\end{aligned}$$

Now it is time to determine whether the difference between the sample proportions is large enough to convince us that $H_0: p_1 = p_2$ is false. To obtain

the sample proportions, we note that, in region, 1, 120 of 1600 were unemployed; so

$$\bar{P}_1 = \frac{120}{1600} = 0.075$$

In region 2, 84 of 1400 were unemployed; so

$$\bar{P}_2 = \frac{84}{1400} = 0.060$$

3. The test statistic is

$$\text{Sample } Z = \frac{\bar{p}_1 - \bar{p}_2}{S_{\bar{p}_1 - \bar{p}_2}}$$

For our data

$$\frac{0.075 - 0.060}{0.00921}$$

$$\text{Sample } z = \frac{0.015}{0.00921} = 1.63$$

Hence, Sample $z = 1.63$

Because $Z = 1.63$ is not greater than the value $Z = 1.96$ specified in the decision rule, we accept H_0 .

4. Accept or reject the H_0

Thus we accept the hypothesis that the proportions unemployed are equal because the samples do not provide convincing evidence that they are different. Figure 5.9 shows that H_0 is accepted because the sample z value, 1.63, lies in the accept H_0 region.

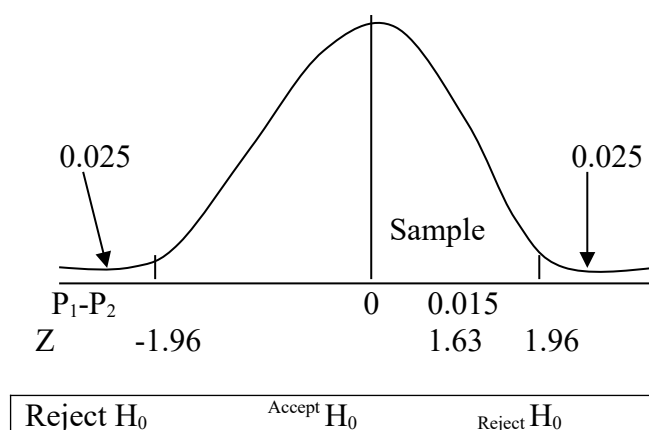


Figure 5.9: Testing the difference between proportions unemployed.

The second option (This one is related with our earlier discussions):

Solution for the same example but different approach

$$H_0: \bar{p}_1 = \bar{p}_2 \text{ or } \bar{p}_1 - \bar{p}_2 = 0 \quad H_1: \bar{p}_1 - \bar{p}_2 \neq 0$$

$$\text{Given } \bar{p}_1 = \frac{120}{1600} = 0.075$$

$$\bar{p}_2 = \frac{84}{1400} = 0.06$$

To get unbiased estimate of population proportion, we combine $\bar{p}_1$ and $\bar{p}_2$

$$\begin{aligned} \text{Pooled estimate } \bar{p} &= \frac{n_1 \bar{p}_1 + n_2 \bar{p}_2}{n_1 + n_2} \\ &= \frac{1600(0.075) + 1400(0.06)}{1600 + 1400} \end{aligned}$$

$$\bar{p} = \frac{120 + 84}{1600 + 1400}$$

$$\bar{p} = \frac{204}{3000} = 0.068 \quad (\text{Which is the same with the previous, first option})$$

$$\text{Now we compute } S_{\bar{p}_1 - \bar{p}_2} = \sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$\begin{aligned} &= \sqrt{0.068(1-0.068)\left(\frac{1}{1600} + \frac{1}{1400}\right)} \\ &= \underline{\underline{0.00921}} \end{aligned}$$

The Z – test statistic is then calculated as:

$$Z = \frac{\bar{p}_1 - \bar{p}_2}{S_{\bar{p}_1 - \bar{p}_2}} = \frac{0.075 - 0.06}{0.00921} = \frac{0.015}{0.00921} = \underline{\underline{1.63}}$$

Since $Z_{cal} (=1.63)$ is less than its table value $z_{\alpha}(-1.96)$, null hypothesis is accepted. Hence, we conclude that the sample proportions unemployed in both regions are equal.

SUMMARY

- Hypothesis testing is a method of using sample information to evaluate a claim that has been converted into a set of statements called the null and alternative hypotheses. The null hypothesis can either be rejected or not rejected; if it is not rejected, the alternative hypothesis, which asserts the null hypothesis to be wrong, must be rejected. The null hypothesis is given the benefit of the doubt and is assumed to be true unless we are faced with statistically overwhelming evidence to the contrary.
- A verbal statement from which a hypothesis test originates may be either directional or non directional. Accordingly, the appropriate hypothesis test will be either directional (one-tail test) or non directional (two-tail test). The object of the hypothesis test may be a single population mean or proportion, or the difference between two population means or proportions.
- When we reject or fail to reject the null hypothesis, there is the possibility that our conclusion may be mistaken. Rejecting a true null hypothesis is known as a Type I error, while failing to reject a false null hypothesis results in a Type II error. Alpha (α), the probability of incorrectly rejecting a true null hypothesis, is also known as the level of significance of the test. Beta (β) is the probability of incorrectly failing to reject a false null hypothesis.
- Hypothesis testing involves (1) formulating the null and alternative hypotheses, (2) selecting the significance level (α) to be used in the test, (3) selecting the test statistic and calculating its value, (4) identifying the critical values for the test statistic and stating the decision rule, (5)

comparing calculated and critical values for the test statistic and reaching a conclusion about the null hypothesis, and (6) making the related business decision for the situation that precipitated the hypothesis test in the first place. In two – tail test, there will be two critical values for the test statistic; in a one –tail test, a single critical value.

- When testing a sample mean, an important consideration in the selection of the test statistic (Z or t) is whether the population standard deviation (σ) is known. If other assumptions are met, the t-test is appropriate whenever σ is unknown.
- If the calculated test statistic falls into a rejection region defined by the critical value(s), the difference between the sample mean or proportion and its hypothesized population value is judged to be too great to have occurred by chance. In such a case, the null hypothesis is rejected.

Activity 5.3

Dear student, try to apply what you have read above on the following questions.

1. In a simple random sample of 600 men taken from a big city, 400 are found to be smokers. In another simple random sample of 900 men taken from another city 450 are smokers. Do the data indicate that there is a significant difference in the habit of smoking in the two cities?
2. Random samples of 500 men and 500 women have been selected to determine whether the proportion of women favoring a political candidate is greater than the proportion of men favoring the candidate. Carry out a hypothesis test at the 1% percent level, if in the samples, 40 women and 20 men favor the candidate.

Exercise 5.3

1. A manufacturer claims that at least 95 percent of the equipments which he supplied to a factory conformed to the specification. An examination of the sample of 200 pieces of equipment revealed that 18 were faulty. Test the claim of the manufacturer.

2. A company is considering two different television advertisements for promotion of a new product. Management believes that advertisement A is more effective than advertisement B. Two test market areas with virtually identical consumer characteristics are selected: Advertisement A is used in one area and advertisement B in the other area. In a random sample of 60 customers who saw advertisement A, 18, had tried the product. In a random sample of 100 customers who saw advertisement B, 22 had tried the product. Does this indicate that advertisement A is more effective than advertisement B, if a 5 percent level of significance is used?